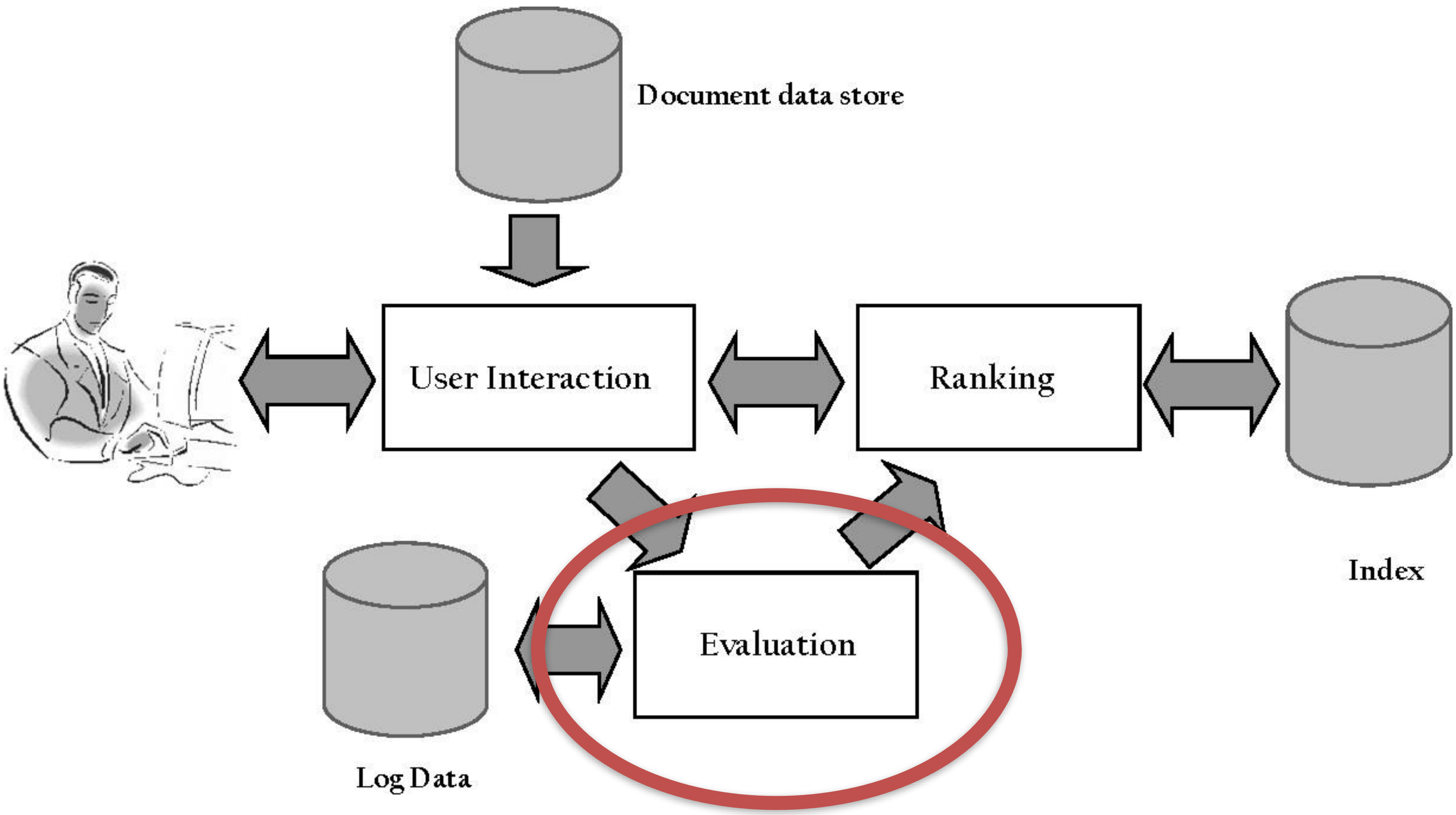
CS418
Lecture 7

Evaluation **measures**

Dr.Ebtsam AbdelHkam

Query Process

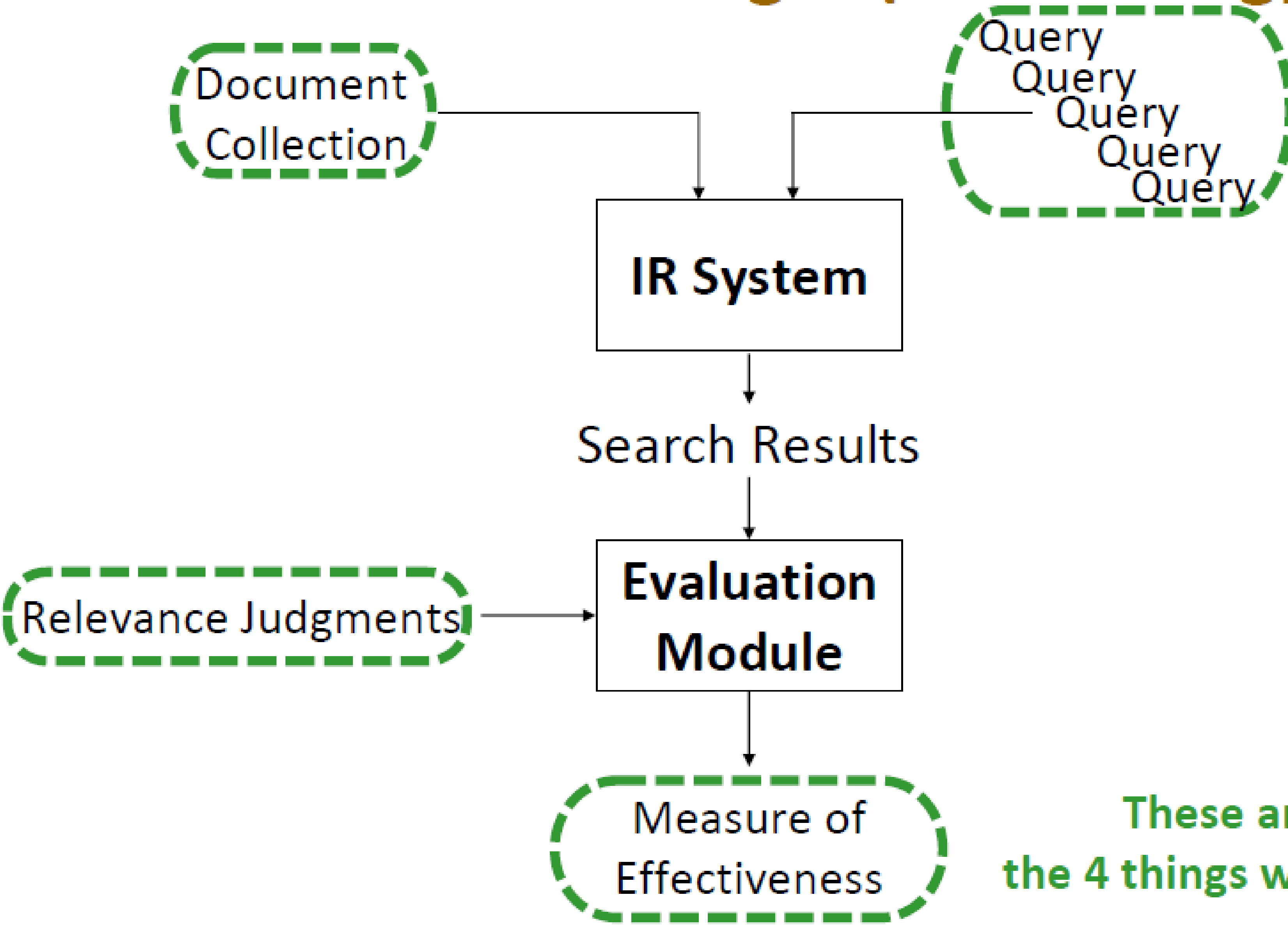


IR Evaluation

- The things we want to evaluate are often subjective, so it's frequently not possible to define a "correct answer."

- Most IR evaluation is comparative: "Is system A or system B better?"

Cranfield Paradigm (Lab setting)



These are the 4 things we need!

Types of Evaluation Strategies

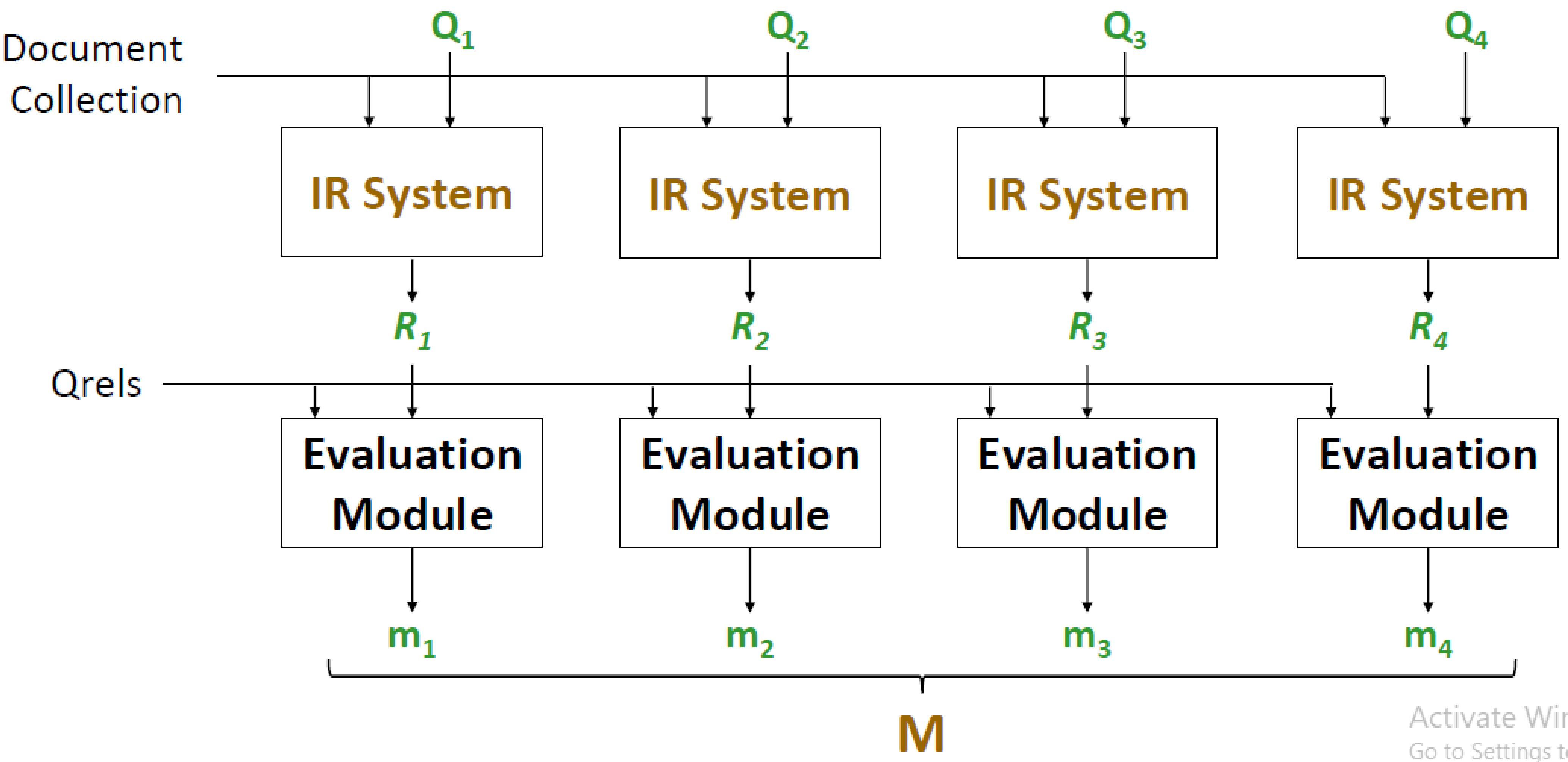
- **System-centered studies**
  - Given documents, queries, and relevance judgments
  - Try several variations of the system
  - Measure which system returns the “best” hit list
  - Laboratory experiment

- **User-centered studies**
  - Given several users, and at least two retrieval systems
  - Have each user try the same task on both systems
  - Measure which system works the “best”

Evaluation Criteria

- Effectiveness
  - How “good” are the documents that are returned?
  - System only, human + system
- Efficiency
  - Retrieval time (query latency/response time), indexing time, index size

Cranfield Paradigm (Lab setting)



Effectiveness Evaluation Measures

- Set-based measures
- Rank-based measures

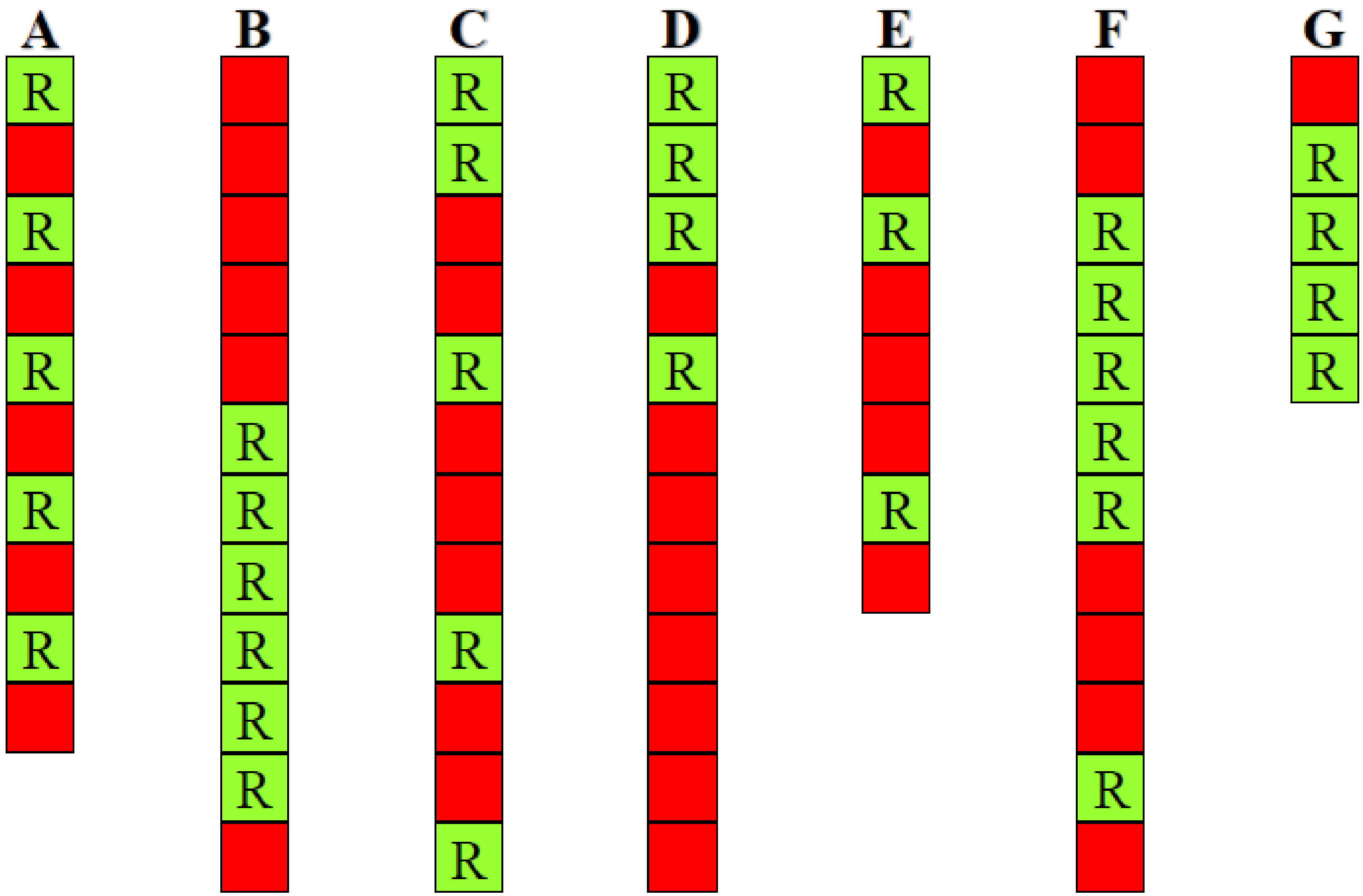
Set-Based Measures

- Assuming IR system returns **<u>set</u>** of retrieved results without ranking.
- No certain number of results per query

- Suitable for Boolean Search

Which is the Best Set?

For **query Q**, **collection has 8 relevant documents** and systems A-G ***retrieved*** following results:



➨ A better list contains *more relevant documents*

But what does “relevant” mean
and how can we measure it?

Set -based measures

Precision and Recall

- **Precision**:

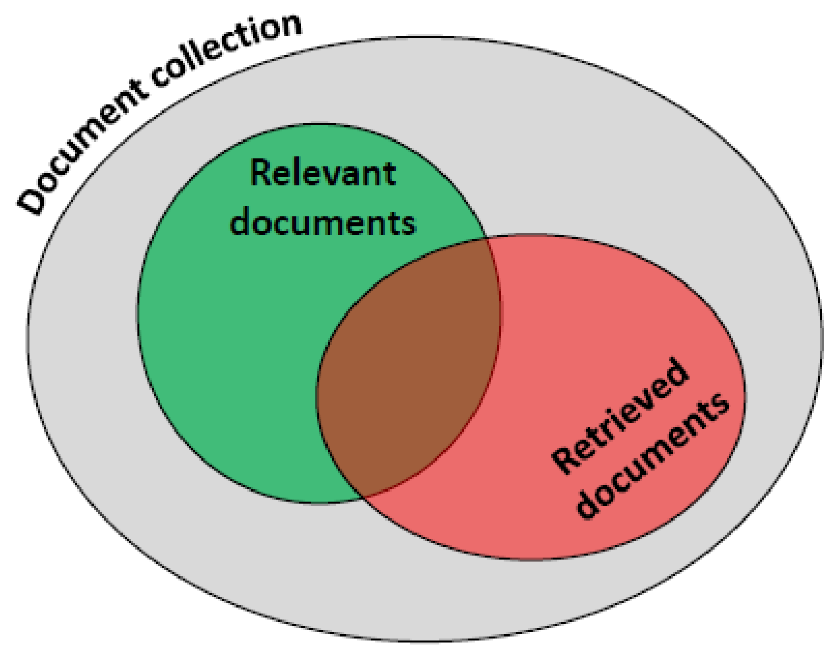
  What fraction of these retrieved docs are relevant?

$$P = \frac{rel \cap ret}{retrieved} = \frac{TP}{TP + FP}$$

- **Recall**:

  What fraction of the relevant docs were retrieved?

$$R = \frac{rel \cap ret}{relevant} = \frac{TP}{TP + FN}$$



	retrieved	not retrieved
irrelevant	FP	TN
relevant	TP	FN

Recall

$$\text{Recall} = \frac{\#(\text{relevant items retrieved})}{\#(\text{relevant items})} = P(\text{retrieved}|\text{relevant})$$

Precision

$$\text{Precision} = \frac{\#(\text{relevant items retrieved})}{\#(\text{retrieved items})} = P(\text{relevant}|\text{retrieved})$$

Recall is the fraction of the relevant documents that are successfully retrieved.

Precision is the fraction of retrieved documents that are relevant to the query

List A

- Relevant
- Non-Relevant
- Non-Relevant
- Relevant
- Non-Relevant

$Relevant = 10$
Retrieved =5

$$Recall = \frac{2}{10}$$

$$Prec = \frac{2}{5}$$

Trade off between P & R

- Precision: The ability to retrieve top-ranked documents that are mostly relevant.
- Recall: The ability to retrieve ***all*** of the relevant items in the corpus.
- Retrieve more docs:
  - Higher chance to find all relevant docs → R ↑↑
  - Higher chance to find more irrelevant docs → P ↓↓

F Measure

The **F Measure** is one way to combine precision and recall into a single value.

- F-measure that combines precision and recall is the harmonic mean of precision and recall, the traditional F-measure or balanced F-score:

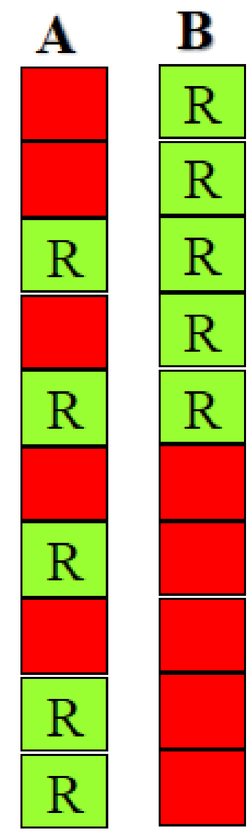
$$\boxed{F = 2 \cdot \frac{\mathrm{precision} \cdot \mathrm{recall}}{\mathrm{precision} + \mathrm{recall}}}$$

- F1 is the *harmonic mean* of precision and recall.

- This heavily penalizes low precision and low recall. Its value is closer to whichever is smaller.

Rank-based IR measures

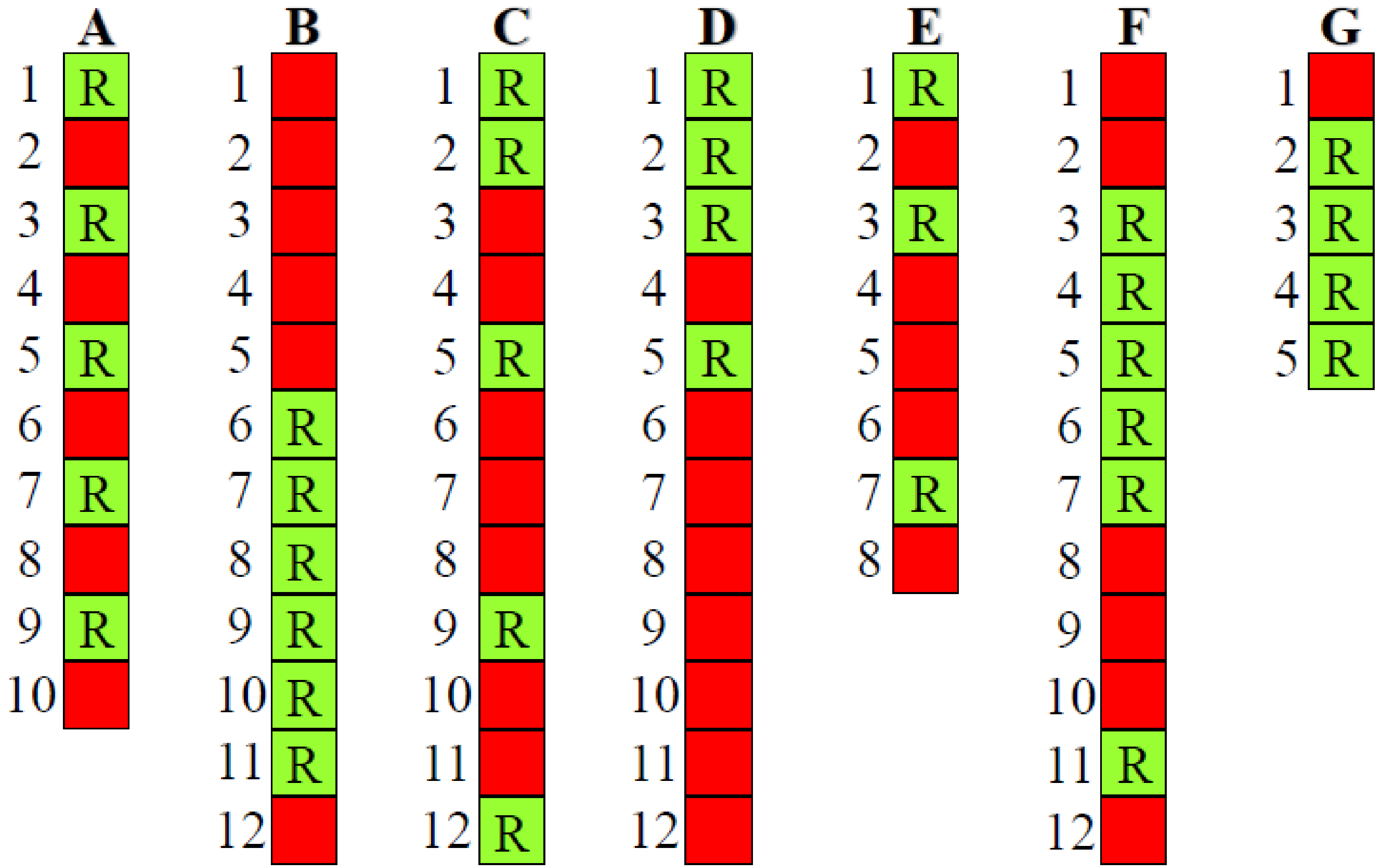
- Consider systems A & B
  - Both retrieved 10 docs, only 5 are relevant
  - P, R & F are the same for both systems
  - Should their performance considered equal?



- Ranked IR requires taking "ranks" into consideration!
- How to do that?

Which is the Best Ranked List?

For **query Q**, **collection has 8 relevant documents** and systems A-G produced following results:



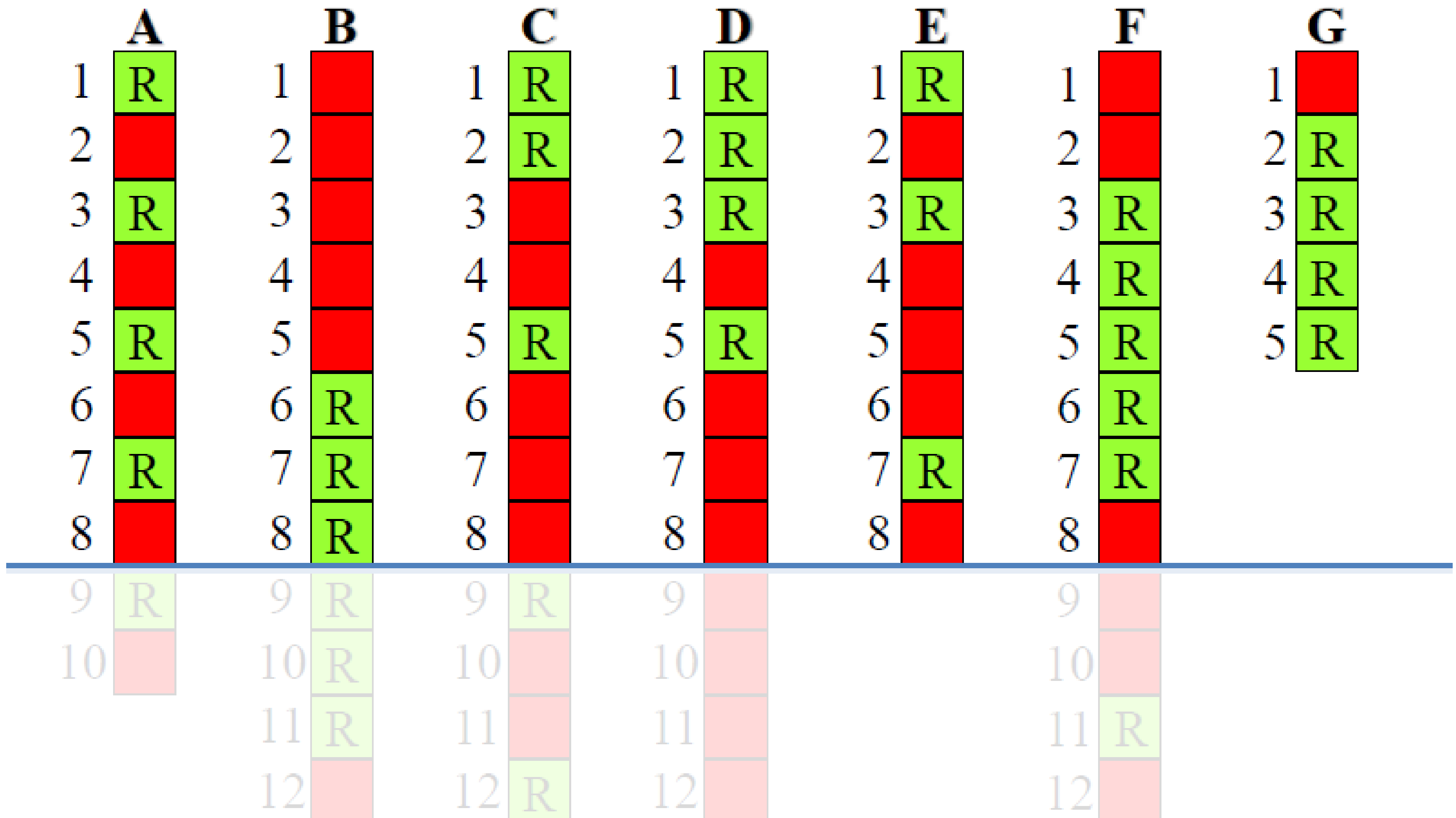
➡ A better list contains *more relevant documents*

➡ A better list has relevant documents ***closer to the top***

Ranked based measures:

Recall@k and P@k

For query Q, **collection has 8 relevant documents** and systems A-G produced following results:



$$\text{Recall} = \frac{\#(\text{relevant items retrieved})}{\#(\text{relevant items})} = P(\text{retrieved}|\text{relevant})$$

$$\text{Precision} = \frac{\#(\text{relevant items retrieved})}{\#(\text{retrieved items})} = P(\text{relevant}|\text{retrieved})$$

Recall@K is almost identical, but truncates your list to the top K elements first.

Precision@K truncates your list to the top K elements.

List A

- Relevant
- Non-Relevant
- Non-Relevant
- Relevant
- Non-Relevant

$Relevant = 10$

Retrieved =5

$$Prec = \frac{2}{5}$$

$$Recall = \frac{2}{10}$$

$$recall@3 = \frac{1}{10}$$

$$Prec@3 = \frac{1}{3}$$

Multiple Queries

Binary Relevance | Graded Relevance | **Multiple Queries**
Test Collections | Ranking for Web Search

Using Multiple Queries

- It isn’t usually fair to compare system performance on a single query. What if the better system just got lucky?
- Instead, we commonly run both systems on a collection of different queries and compare metric values across all queries.

➡ Individual queries can still be useful. Look for distinctive queries: a system’s best or worst query, the queries for which the overall worse system beats the overall better system, etc.

AP & MAP measures

- One common way to combine information across queries is simply to take the mean of the metric over the queries.

- **Mean Average Precision** (MAP) is the average AP value for a system across many queries.

  ➡ This is one of the most popular evaluation metrics when using binary relevance.

AP & MAP formally

$$AP = \frac{1}{r}\sum_{k=1}^{n} P@k \times rel(k)$$

r: number of relevant docs for a given query

n: number of documents retrieved

$rel(k)$: 1 if retrieved doc @ k is relevant, 0 otherwise.

$$MAP = \frac{1}{Q}\sum_{q=1}^{Q} AP(q)$$

Q: number of queries in the test collection

Average Precision (AP)

Q_1 (has 5 rel. docs)

Rank	Rel.	Precision
1	R	1/1=1.00
2	R	2/2=1.00
3		
4		
5	R	3/5=0.60
6		
7		
8		
9	R	4/9=0.44
10		

AP = 3.04 / 5
= **0.608**

Q_2 (has 3 rel. docs)

Rank	Rel.	Precision
1		
2		
3	R	1/3=0.33
4		
5		
6		
7	R	2/7=0.29
8		
⋮		$\frac{3}{\infty}=0$

AP = 0.62 / 3
= **0.207**

Q_3 (has 7 rel. docs)

Rank	Rel.	Precision
1		
2	R	1/2=0.50
3		
4		
5	R	2/5=0.40
6		
7		
8	R	3/8=0.375
9		
⋮		

AP = 1.275 / 7
= **0.182**

Mean Average Precision (MAP)

Rank	Q_1 (has 5 rel. docs)		Q_2 (has 3 rel. docs)		Q_3 (has 7 rel. docs)	
1	R	1/1=1.00				
2	R	2/2=1.00			R	1/2=0.50
3			R	1/3=0.33		
4						
5	R	3/5=0.60			R	2/5=0.40
6						
7			R	2/7=0.29		
8					R	3/8=0.375
9	R	4/9=0.44				
10						
	AP = 0.608		AP = 0.207		AP = 0.182	

MAP = (0.608+0.207+0.182)/3 = **0.332**

8. An IR system produces the following rankings in answer to queries q_1 and q_2. The underscored documents are the ones relevant to the user.

Compute the Mean Average Precision

R	q_1	q_2
1	$\underline{\text{A}}$	$\underline{\text{F}}$
2	L	$\underline{\text{G}}$
3	$\underline{\text{G}}$	D
4	F	$\underline{\text{E}}$
5	D	L
6	$\underline{\text{E}}$	I
7	$\underline{\text{B}}$	H
8	$\underline{\text{H}}$	C
9	I	$\underline{\text{B}}$
10	C	A

R	q_1	q_2
1	A	F
2	L	G
3	G	D
4	F	E
5	D	L
6	E	I
7	B	H
8	H	C
9	I	B
10	C	A

Retrieved documents	P_{q_1}		P_{q_2}
1	1/1		1/1
2			2/2
3	2/3		
4			3/4
5			
6	3/6		
7	4/7		
8	5/8		
9			4/9
10			

$$AP_{q_1} = \frac{1/1 + 2/3 + 3/6 + 4/7 + 5/8}{5} = 0.67$$

$$AP_{q_2} = \frac{1/1 + 2/2 + 3/4 + 4/9}{4} = 0.80$$

$$MAP = \frac{0.67 + 0.80}{2} = 0.74$$